WILP, BITS Pilani

MACHINE LEARNING · COMPANION READER

Advanced Decision Trees & Pruning

Continuous Attributes, Overfitting, Pre/Post-Pruning, Gini Index, Gain Ratio, and MDL

Session 8 · Read alongside the lecture slides · Every slide example worked out in full

Prepared for the Work Integrated Learning Programmes (WILP), BITS Pilani.

1 Continuous Attributes in Decision Trees

Continuous features are converted into binary decision thresholds by sorting values and testing candidate midpoints.

Given sorted values $a_1 < a_2 < \cdots < a_N$, candidate midpoints $c_i = (a_i + a_{i+1})/2$ are evaluated for maximum Information Gain.

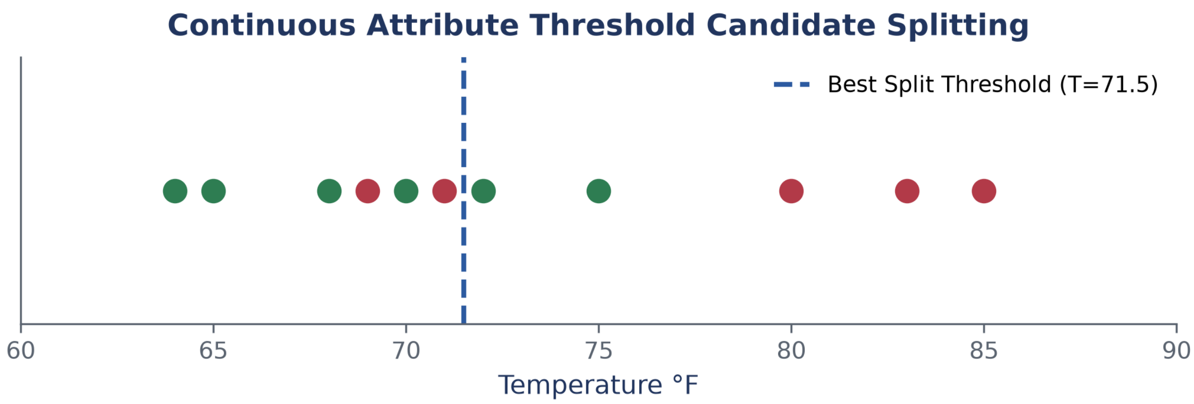

Figure 1: Candidate split thresholds for a continuous attribute.

2 Model Overfitting & Tree Pruning

Overfitting occurs when a tree memorizes training noise at the expense of generalization accuracy.

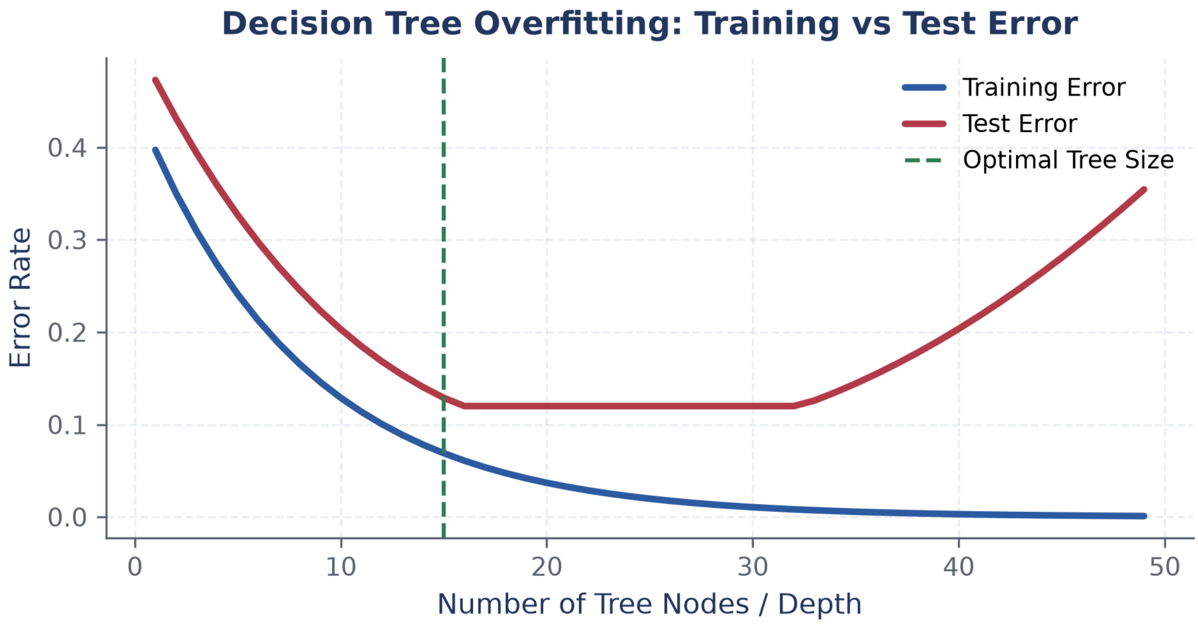

Figure 2: Training versus test error as decision tree complexity increases.

2.1 Pruning Strategies

1. **Pre-Pruning:** Halt node expansion early if Information Gain is below a threshold.
2. **Post-Pruning:** Trim subtrees from a fully grown tree using validation data.

3 Alternative Impurity Measures: Gini & Gain Ratio

Gini Index and Gain Ratio address the limitations of standard Information Gain.

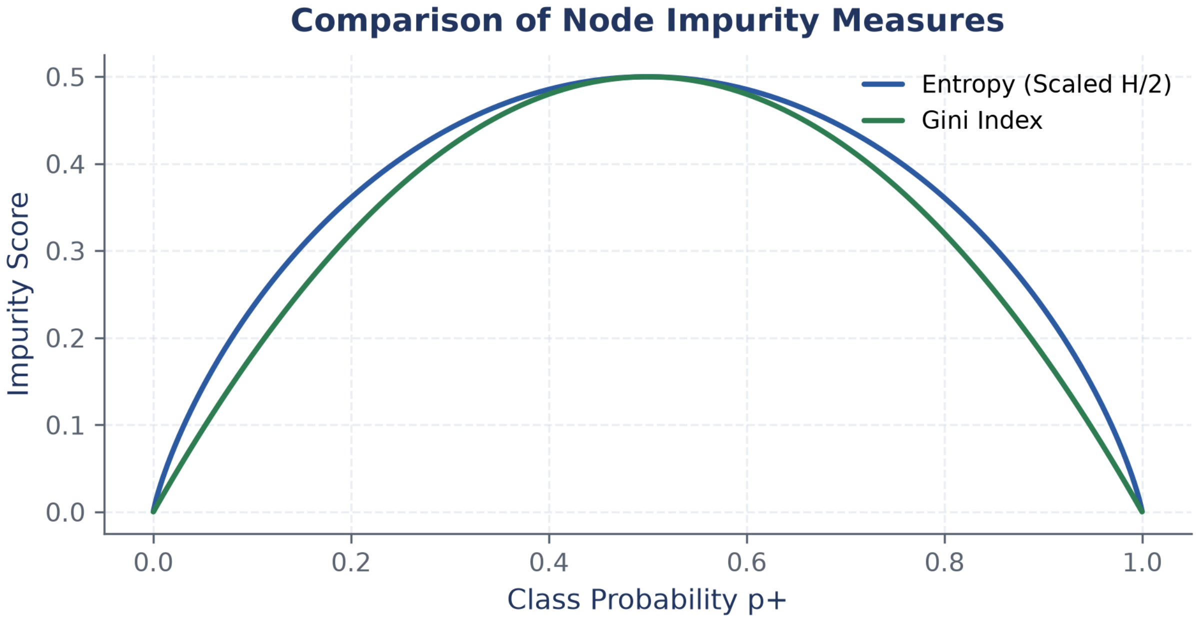

Figure 3: Comparison of impurity metrics: Gini Index vs Scaled Entropy.

$$\mathrm{Gini}(S) = 1 - \sum_{i=1}^{c} p_i^2, \quad \mathrm{GainRatio}(S, A) = \frac{\mathrm{Gain}(S, A)}{\mathrm{SplitInfo}(S, A)}. \tag{1}$$

4 Minimum Description Length (MDL) Principle

MDL balances model complexity against training error.

$$\text{Cost}(\text{Model}, \text{Data}) = \text{Cost}(\text{Model}) + \text{Cost}(\text{Data} \mid \text{Model}). \tag{2}$$